Lecture 10

1. The Inverse of the Matrix

Def: We say and $n \times n$ matrix is invertible if there is another $n \times n$ matrix which we will call A^{-1} such that

$$A^{-1}A = AA^{-1} = I_n$$

A^{-1}: The inverse of A

inverse is unique

inverse doesn't exist = singular

The: Let A be a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$\det(A) = ad - bc$ determinant of A

A is invertible $\iff$ $\det(A) \neq 0$

If $\det(A) \neq 0$, then

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The: If A is an invertible $n \times n$ matrix, then for each $\vec{b} \in \mathbb{R}^n$, the equation:

$$A\vec{x} = \vec{b}$$

has a unique solution

$$\vec{x} = A^{-1}\vec{b}$$

2. Properties of the Inverse

Let A, B be invertible $n \times n$ matrices, then:

(1) A^{-1} is invertible and $(A^{-1})^{-1} = A$

(2) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$

(3) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$

(4) $A_1, \dots, A_k$ is invertible and $A_1 A_2 \cdots A_{k-1} A_k$ is invertible and

$$(A_1 A_2 \cdots A_{k-1} A_k)^{-1} = A_k^{-1} A_{k-1}^{-1} \cdots A_2^{-1} A_1^{-1}$$

3. How to find the inverse of a matrix

Define E_i is the ith elementary row operation

$$\begin{cases} A \longrightarrow E_1 A \longrightarrow E_2 E_1 A \longrightarrow \cdots \longrightarrow E_k E_{k-1} \cdots E_2 E_1 A = I \\ I \longrightarrow E_1 I \longrightarrow E_2 E_1 I \longrightarrow \cdots \longrightarrow E_k E_{k-1} \cdots E_2 E_1 I = A^{-1} \end{cases}$$

($E_k E_{k-1} \cdots E_2 E_1$ is the inverse of A)

$\therefore$ $[A|I] \xrightarrow{ERO} [I|A^{-1}]$

Thm: $A \in \mathbb{R}^{n\times n}$ is invertible if and only if it is row equivalent to identity matrix

4. Characterizations of Invertible Matrix

Let A be an $n \times n$ matrix, then the following statements are equivalent:

1. A is invertible
2. A is row equivalent to the identity
3. A has n pivot positions
4. The equation $A\vec{x} = 0$ has only the trivial solution
5. The columns of A are linearly independent
6. The linear transformation $T(\vec{x}) = A\vec{x}$ is one-to-one
7. The equation $A\vec{x} = \vec{b}$ has at least one solution for each $\vec{b}$ in $\mathbb{R}^n$
8. The columns of A span $\mathbb{R}^n$
9. The linear transformation $T(\vec{x}) = A\vec{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$
10. There is an $n \times n$ matrix C such that $CA = I$
11. There is an $n \times n$ matrix D such that $AD = I$
12. A^T is an invertible matrix

5. Invertible Linear Transformation

Def: A linear transformation $T: \mathbb{R}^n \to \mathbb{R}^n$ is invertible if we can find the linear transformation T^{-1} such that

$$T^{-1}(T(\vec{x})) = T(T^{-1}(x)) = \vec{x}$$

for all $x \in \mathbb{R}^n$

T^{-1} is unique

Thm: $T: \mathbb{R}^n \to \mathbb{R}^n$ is invertible

$\Leftrightarrow$ standard matrix is invertible

T^{-1} is the matrix transformation of A^{-1}

$$T^{-1}(\vec{x}) = A^{-1}\vec{x}$$